Yoligi Harshini

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.45
11007-SRI VIVEKANANDA JR COLLEGE	Class XII (INTERMEDIATE EDUCATION)	2023	97.80%
ST AUGUSTINE ENGLISH MEDIUM HIGH SCHOOL	Class X (SECONDARY EDUCATION)	2021	98.10%

EXPERIENCE

Undergraduate Researcher

Enhancing computation of Betweenness Centrality for Graphs

- **Engineered** parallel BC algorithm with source-sampling and cost-aware scheduling, achieving **874× speedup.**
- **Designed** cost-aware scheduling by component size and degree, achieving **48-136× speedup** over parallel exact.
- **Benchmarked** on Twitter and G+ SNAP datasets (up to **13.6M edges**), achieving **91-95% top-20 accuracy.**

SKILLS

Programming Languages: C, C++, RISC-V.

Web Technologies: HTML5, CSS3, MySQL, MongoDB, FastAPI.

Tools: LaTeX, Git and GitHub, Solid Edge.

Operating Systems: Linux (Ubuntu), Windows.

Communication Skills: Telugu, English, Kannada.

PROJECTS

Notification Framework | Software Engineering

- **Engineered** multi-channel notification platform supporting **4 channels** with asynchronous Celery/RabbitMQ processing.
- **Implemented 3-layer rate limiting** at **5,000 req/min** system capacity with automatic retries and DLQ-based recovery.
- **Developed** observability dashboard tracking **p50/p95/p99 latency**, delivery status, queue depth, and webhook activity.

Multi-Client Quiz System | Computer Networks

- **Engineered** multi-client TCP quiz system supporting **10 concurrent clients** with threaded client-server communication.
- **Achieved 718.5 Bytes/s throughput** across **7 clients**, up from **208.7 Bytes/s** with a single client.
- **Measured 194-1568 µs latency** across **1-7 clients**, analyzing scalability bottlenecks from threading and lock contention.

RISC-V Assembler | Computer Architecture

- **Constructed a RISC-V** assembler in **C++** to convert assembly instructions into corresponding **hexadecimal**

machine code.
- **Integrated** support for **R, I,** and **S**-type instructions of size **32 bit,** ensuring accurate parsing and translation.
- **Simulated instruction-level behaviour of the RISC-V ISA**, deepening understanding of computer architecture.

Sudoku Validator | Operating Systems

- **Structured a multithreaded** program using **pthreads** to **validate Sudoku** of size upto **100x100** for correctness in **parallel**.
- **Applied** chunk, mixed, and sequential patterns for concurrent rows, columns, subgrids, achieving **20% faster execution**.
- **Integrated** custom **mutual exclusion algorithms--TAS, CAS** --to ensure thread-safe validation.

Infinite-Precision Arithmetic | Software Development Fundamentals

- **Designed** and **developed** a C++ calculator to perform **addition** and **subtraction** on **integers exceeding 10^9 digits.**
- **Handled infinite-range numbers** by implementing **custom data structures,** ensuring functions beyond standard data types.
- **Collaborated** in a **2-member team,** focusing on **memory-efficient logic** to support scalability and reliability.

Vehicle Rental System | Database Management System

- **Crafted a full-stack web application** to enable **online rental of 2-wheelers** and **4-wheelers** for customizable durations.
- **Cooperated** in a **4-member team** using **Django** and **MySQL** for backend development and data management.
- **Enhanced responsive user interfaces** using **HTML** and **CSS**, ensuring smooth user interaction and booking experience.

RELEVANT COURSES

- **Computer Science:** Data Structures, Algorithms, Operating Systems, Database Management System, Computer Networks, Software Development Fundamentals, Computer Architecture, Software Engineering, Compilers, Discrete Mathematics, Theory of Computation, Foundations of Machine Learning.

- **Other:** Probability and Random Variables, Calculus, Digital Circuits.

POSITIONS OF RESPONSIBILITY

International Relations Cell

- **Worked** as a **Web Development Coordinator** for the IR Cell, contributing to website development and maintenance.
- **Collaborated with the core team** in planning technical and design updates for the IRC website and event portals.

EXTRACURRICULAR

Sports and Games: Actively participate in **badminton** sessions and enjoy indoor games such as **Carroms** and **UNO cards.**

National Service Scheme Volunteer, IITH: Volunteered with the **NSS**, contributing **120+** hours to social service initiatives.

- Participated in various **community service** activities and contributed to meaningful **social outreach** programs.

Hobbies and Interests: Exploring a wide range of movies, dramas, and web series across diverse genres and cultures.